

usage as one tends to install and load numerous applications. These applications, even after being removed, leave behind a garbage of stray entries in the system registry, leading to fragmentation of hard disk data and in some cases, lost clusters are formed on the hard disk due to repeated crashes or faulty uninstalls. Specific tools in a system suite help prevent these problems and even correct them if something goes wrong.

We first tested the defragmentation tools bundled with the system utilities. The DiskTune utility in the McAfee Utilities suite performed admirably and left the competition behind in this comparison. It took just 12 minutes 48 seconds to defragment 1.7 GB of data on the test hard disk. This is one of the fastest times logged by any suite we tested, allowing you to effortlessly defragment your hard disk as often as once a week. Norton



Defragmenting a hard disk takes a lot of time. Start the process just before you go off to bed or when you leave office for home

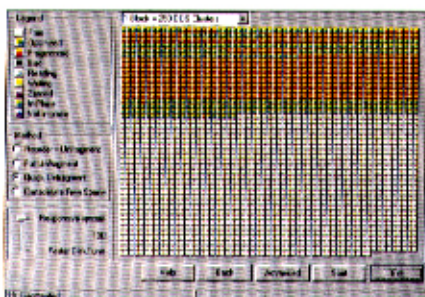
want of space. The program can then be restored with all its settings intact, whenever desired. The best thing about this utility is that you need not install it—it can be run from the CD itself, saving precious hard disk space.

Next up was the test of the registry cleaners. To evaluate their effectiveness, we directly deleted the folders containing Microsoft Office XP, Adobe Photoshop 6.0, Cyberlink PowerDVD and Roxio EasyCD Creator from the test hard disk, instead of taking the recommended route of uninstalling them first. Thus, stray registry entries of up to 0.7 MB were created—these strays were then used to test our suites.

Norton Utilities did find a rather large list of stray entries, but there was no automated deletion process. You will thus be forced to delete each entry manually—this could be a very time consuming task. As such, the interface does not allow you to select an entire registry set, so that it can be cleaned at one go. Hence, you would be forced to regularly check for stray registries, since if the list gets long, the task of cleaning becomes an ordeal. This is a huge drawback, hence we didn't give Norton any points in this test. McAfee offers a wizard to help clean the registry and at the end of the process it presents the old and new registry size and number of registry entries cleaned. However, the system was unstable after the registry was optimised using the McAfee Registry Wizard.

System Recovery: System crashes are like changes in the weather—totally unpredictable and inevitable. There is very little one can do about a crash apart from figure out a means to minimise its effects. Crash protection tools are primarily meant to avoid a system crash. They come into action when the application detects a level of instability in the operating system, usually due to a program error (just before the dreaded blue screen or a page fault error). They offer

you some time to save valuable data after this frustrating event, so that files are not lost. Crash-Protector and CrashProof from McAfee and Ontrack, respectively, effectively guard you against any OS misdeemeanours. Surprisingly, Norton disappointed us by not bundling an equivalent utility



Defragmentation utilities such as McAfee DiskTune shown here, help increase system performance by reducing disk access time

SystemWorks 2002 performed poorly and logged the highest time—21 minutes 39 seconds. Thus, a 20 GB hard disk will take more than 3 hours to be defragmented using the Norton suite. The JetDefrag utility in the Ontrack suite managed to complete the task in 15 minutes 9 seconds—a very respectable time.

Uninstall utilities are an extension to the Add/Remove utility which is found by default in Windows. CleanSweep, bundled with Norton suite, is the only utility that removes installed programs, and also helps clean up leftover Internet files and other temporary and residual data on the hard disk. This utility can also back up uninstalled programs if you so desire—a nifty feature if you are forced to remove a program for



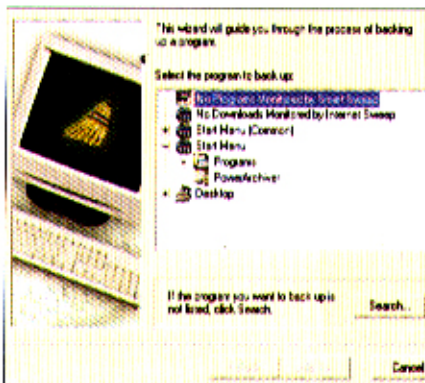
Use a permanent data-erasing tool to delete critical or confidential files; but use the tool wisely

Defrag Demystified

Data is stored on a hard disk in chunks called sectors. To access a file, a sector is read by a searching mechanical arm that houses a read/write head. For optimal disk performance, this searching motion should be minimal. To ensure this, it is imperative that data files be stored in a sequential manner. However, over time, deleting files, installing and reinstalling software causes the data sectors to be scattered over the surface of the disk in a non-contiguous manner, leading to what is commonly termed as 'disk fragmentation'. The result is a predictable increase in the time taken to access data, directly leading to a decrease in system performance. This is where a 'defrag' utility steps in—it functions by gathering the scattered data sectors into contiguous and disk-friendly chunks. A good defragmenter may attempt to optimise things further by placing application and operating system data close to each other, thus minimising disk movement, decreasing data access time and shaving load times off an application. Additionally, a defragmented disk makes it easier to retrieve lost data.

that accomplished this level of functionality.

A bootable disk is another great feature that lets you start the system despite a hard disk failure. This is done via the software CD or by creating a rescue disk.



Norton CleanSweep backs up program settings and system files

A disk of this nature lets you boot the system and restore it to a safe state. Roxio GoBack 3 in the Ontrack suite and SystemSaver from Norton are system restore utilities that provide this functionality. McAfee, sadly, did not bundle any such utility.